

Experiment-4

Aim: Install C Compiler in VirtualBox VM (Cloudera) and run C programs.

Description

Step 1 — Start the Virtual Machine

1. Open **Oracle VM VirtualBox Manager**.
2. Select your VM (Cloudera / Ubuntu / Debian).
3. Click **Start → Normal Start**.
4. Wait for the OS to boot and log in.

Step 2 — Update the OS (Important)

Open a terminal inside the VM:

```
sudo apt update
```

```
sudo apt upgrade -y
```

Purpose of the above commands:

- `sudo` – Runs command with administrator privileges
- `apt` – Package management tool
- `update` – Refreshes package list
- `upgrade` – Upgrades installed packages

Step 3 — Install GCC (GNU C Compiler)

```
sudo apt install build-essential -y
```

```
gcc --version
```

This installs:

- `gcc`
- `g++`
- `make`
- header files

Step 4 — Create Directory for C Programs

```
mkdir ~/24485A0503
```

```
cd 24485A0503
```

Step 5 —

Program 1: Hello world program



```

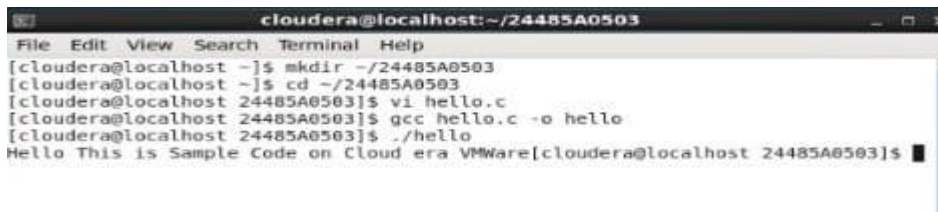
cloudera@localhost:~/24485A0503
File Edit View Search Terminal Help
#include<stdio.h>
void main()
{
    printf("Hello This is Sample Code on Cloud era VMWare");
}

```

Save:

- Press i → Enter
- Press ESC
- Type :wq → Enter

Compile & Run:



```

cloudera@localhost:~/24485A0503
[cloudera@localhost ~]$ mkdir ~/24485A0503
[cloudera@localhost ~]$ cd ~/24485A0503
[cloudera@localhost 24485A0503]$ vi hello.c
[cloudera@localhost 24485A0503]$ gcc hello.c -o hello
[cloudera@localhost 24485A0503]$ ./hello
Hello This is Sample Code on Cloud era VMWare[cloudera@localhost 24485A0503]$

```

Program 2: Sum of 2 numbers



```

cloudera@localhost:~/24485A0503
File Edit View Search Terminal Help
#include<stdio.h>
void main()
{
    int a,b,sum;
    printf("Enter a,b values:");
    scanf("%d %d",&a,&b);
    sum=a+b;
    printf("The sum of a and b is %d\n",sum);
}

```

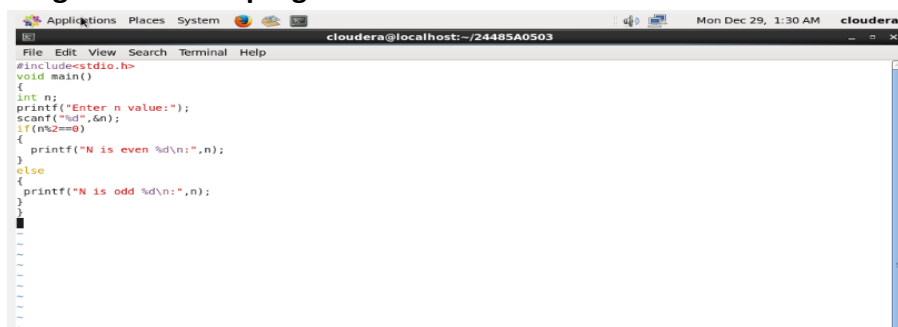
Compile & Run:

```

[cloudera@localhost 24485A0503]$ vi sum.c
[cloudera@localhost 24485A0503]$ gcc sum.c -o sum
[cloudera@localhost 24485A0503]$ ./sum
Enter a,b values:10 20
The sum of a and b is 30
[cloudera@localhost 24485A0503]$

```

Program 3: if else program



```

cloudera@localhost:~/24485A0503
File Edit View Search Terminal Help
#include<stdio.h>
void main()
{
    int n;
    printf("Enter n value:");
    scanf("%d",&n);
    if(n%2==0)
    {
        printf("N is even %d\n",n);
    }
    else
    {
        printf("N is odd %d\n",n);
    }
}

```

Compile & Run:

```
[cloudera@localhost 24485A0503]$ vi condi.c
[cloudera@localhost 24485A0503]$ gcc condi.c -o condi
[cloudera@localhost 24485A0503]$ ./condi
Enter n value:5
N is odd 5
:[cloudera@localhost 24485A0503]$
```

Program 4: Factorial of a Number

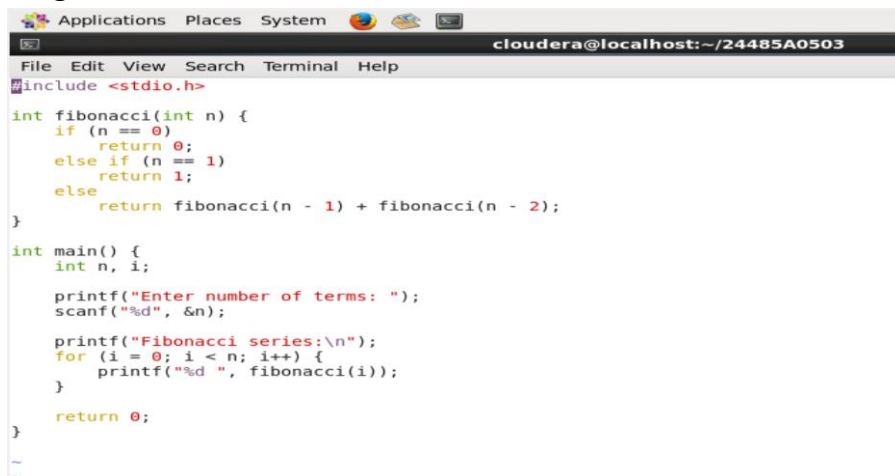

```

Applications  Places  System  cloudera@localhost:~/24485A0503
File Edit View Search Terminal Help
#include<stdio.h>
void main()
{
    int n,i=1,fact=1;
    printf("Enter n value:");
    scanf("%d",&n);
    while(i<=n)
    {
        fact=fact*i;
        i++;
    }
    printf("Factorial of %d is %d\n",n,fact);
}

```

Compile & Run:

```
[cloudera@localhost 24485A0503]$ vi fact.c
[cloudera@localhost 24485A0503]$ gcc fact.c -o fact
[cloudera@localhost 24485A0503]$ ./fact
Enter n value:5
Factorial of 5 is 120
[cloudera@localhost 24485A0503]$
```

Program 5: Fibonacci of a number


```

Applications  Places  System  cloudera@localhost:~/24485A0503
File Edit View Search Terminal Help
#include <stdio.h>
int fibonacci(int n) {
    if (n == 0)
        return 0;
    else if (n == 1)
        return 1;
    else
        return fibonacci(n - 1) + fibonacci(n - 2);
}
int main() {
    int n, i;
    printf("Enter number of terms: ");
    scanf("%d", &n);
    printf("Fibonacci series:\n");
    for (i = 0; i < n; i++) {
        printf("%d ", fibonacci(i));
    }
    return 0;
}

```

Compile & Run:

```
[cloudera@localhost 24485A0503]$ vi fib.c
[cloudera@localhost 24485A0503]$ gcc fib.c -o fib
[cloudera@localhost 24485A0503]$ ./fib
Enter number of terms: 10
Fibonacci series:
0 1 1 2 3 5 8 13 21 34 [cloudera@localhost 24485A0503]$
```

**Program 6 : pattern (*)**

```

cloudera@localhost:~/24485A0503
#include <stdio.h>
void main()
{
    int i,j,n;
    printf("Enter no of rows:");
    scanf("%d",&n);
    for(i=1; i<=n; i++)
    {
        for(j=1; j<=n-i; j++)
        {
            printf(" ");
        }
        for(j=1; j<=2*i-1; j++)
        {
            printf("*");
        }
        printf("\n");
    }
}

```

Compile & Run:

```

[cloudera@localhost 24485A0503]$ vi for.c
[cloudera@localhost 24485A0503]$ gcc for.c -o for
[cloudera@localhost 24485A0503]$ ./for
Enter no of rows:5
    *
   ***
  *****
 *****
*****

```

Program 7: Creating a process using OS

```

cloudera@localhost:~/24485A0503
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>

int main()
{
    pid_t pid;
    pid = fork(); // System Call

    if (pid < 0) {
        fprintf(stderr, "Fork Failed");
        return 1;
    } else if (pid == 0) {
        printf("I am the Child. My PID is %d\n", getpid());
    } else {
        printf("I am the Parent. My Child's PID is %d\n", pid);
        return 0;
    }
}

```

Compile & Run:

```

[cloudera@localhost ~]$ mkdir 24485A0503
[cloudera@localhost ~]$ cd 24485A0503
[cloudera@localhost 24485A0503]$ vi os.c
[cloudera@localhost 24485A0503]$ gcc os.c -o os
[cloudera@localhost 24485A0503]$ ./os
I am the Parent. My Child's PID is 11926
[cloudera@localhost 24485A0503]$ I am the Child. My PID is 11926
[cloudera@localhost 24485A0503]$

```